

Combined Effect of Progressive Muscle Relaxation, Autogenic Training and Meditation Practices on Mental Toughness, Self Confidence, Stress and Anxiety among Athletes

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ABSTRACT

The purpose of the study was to find out the combined effect of relaxation training including progressive muscle relaxation, autogenic training and meditation practices on mental toughness, self confidence, stress and anxiety among athletes. For achieving the purpose of the study total of 30 men athletes were selected as samples from the Athletic academy. All the subjects are with minimum of inter-collegiate level athletic participation and studying in arts and science courses; and taking training in the athletic academy. Their age group from 19 to 24 years and the selected subjects were divided into two groups equally with 15 each, as control and experimental group. The experimental group underwent combination of relaxation training including progressive relaxation, autogenic training and meditation practices for a duration of one hour in an order of all three training with twenty minutes duration each for the period of eight weeks in a schedule of weekly six days with one hour session each. Analyze of Covariance (ANCOVA) was used as a statistical tool. The statistical findings of the study revealed that the experimental group done the combination of relaxation training includes progressive relaxation; autogenic training and meditation practices significantly improved the mental toughness and self-confidence; and reduced stress and anxiety levels of athletes.

KEYWORDS

Mental Toughness, Self-Confidence, Stress, Anxiety, Progressive Muscle Relaxation, Autogenic Training, Meditation.

INTRODUCTION

Relaxation procedures have emerged as one of the training to improve psychological well-being and sports excellence. It is potentially useful to improve physiological and psychological qualities. Relaxation procedures are producing the body's natural relaxation response and

increased well-being. Stress and anxiety as well as various health problems commonly associated with relaxation techniques. Relaxation is great important to any athlete striving for peak performance. It is a method, process, procedures or activity that helps to relax.

Relaxation can be achieved through meditation, autogenic and

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progressive muscle relaxation. Relaxation helps to improve coping with stress. Stress is the leading cause of mental and physical problems; therefore, it is essential to follow systematic relaxation procedure to enhance the psychological benefits. According to the Oxford dictionary, relaxation is when the body and mind are free from tension and anxiety.

Progressive muscle relaxation technique was introduced by Edmond Jacobson (1938). It is a two-step process in which is systematically tense and relaxes different muscle groups in the body. This technique will make more aware of the difference between the relaxation and tension of muscles: this is a way to become more aware of your physical sensations.

Autogenic training is a technique of self-hypnosis developed by Dr H.H. Sultz, a German neurologist. Autogenic training is defined as a psychophysiological self-control technique aiming at physical and mental relaxation (Schultz & Luthe, 1969). The technique consists of a series of six mental exercises used to elicit the bodily sensations of warmth and heaviness. It is a relaxation technique utilizes both body awareness and visual imagery in order to release tension and stress.

Meditation is a practice where an individual uses a technique, such as mindfulness, or focusing their mind on a particular object, thought or activity and to train attention and awareness, and achieve a mentally clear and emotionally calm and stable state (Jevning, 1992). It is a form of relaxation and directly connected with

selective attention, which will enhance many psychological qualities. Meditation will result in the relaxation response if practiced in a quiet environment that is associated with a passive attitude and decreased muscle tone (Benson et al, 1974).

All types of relaxation strategy involve use of the self-regulation of strength. This strength involves taking control of breathing and creating a greater feeling of calmness in body's physiology. In turn, such self-regulation has a positive impact on the racing of mind and worries of the day and these are ultimately brings many bodily changes in physiologically and psychologically. Such wonderful relaxation strategies are to be included in the training regime.

There are number of studies are conducted on the effects of relaxation training on different psychological qualities and all the studies were given a positive result. Navaneethan and Soundara Rajan (2010) found that the six weeks progressive muscle relaxation training had significant difference in competitive anxiety among the male inter-collegiate Volleyball players. Bulent Okan Micoogullari (2013) examined the impact of six weeks psychological skills training and the result shown the improvement in self-confidence among Basketball Players. Shapiro and Lehrer (1980) found five weeks autogenic training and progressive relaxation significantly decreased the anxiety. Syed Anwar Ali and Asath Ali Khan (2018) found eight weeks autogenic training were beneficial effect on stress and anxiety

among Handball players. Dhanaraj (2018) investigated that eight weeks autogenic training had significant improvement on sports competition anxiety and self-confidence among college Cricket players. Seung-Joo Lim, Chunmi Kim (2014) found eight weeks autogenic training beneficial for stress reduction in nursing students. Nitin Sakhar and Kalpana (2018) investigated that the relaxation technique is an effective intervention for reducing competitive state anxiety and improving self-confidence.

The present study was undertaken to identify the different psychological benefits from the combined benefits of three relaxation training of deep muscle relaxation, autogenic training and meditation on the selected psychological variables of mental toughness, self-confidence stress and anxiety.

METHODOLOGY

The selection of subjects, training procedure, testing procedure and statistical procedure were explained below.

Selection of subjects

For achieving the purpose of the study total of 30 subjects were selected as samples from Athletic academy from Chennai city. Their age group between 19 to 24 years and the selected subjects were divided into two groups equally with 15 each as control and experimental group. All the selected subjects were with minimum of inter collegiate athletic participation and studying in an Arts and Science College; and taking training in the Athletic academy.

Training Procedure

The experimental group underwent combination of relaxation Training which includes progressive relaxation, autogenic training and meditation practices for a duration of one hour in an order of all three training with twenty minutes duration each for the period of eight weeks in a schedule of weekly six days with one hour session each.

Testing procedure

The pre and posttest were conducted on selected psychological variables of mental toughness by using Alan Goldberg Mental Toughness Questionnaire (MTQ). Mental toughness questionnaire consists of 30 statements. Each answer has one mark and the statement having true as well as false answer. The overall score of 26-30 indicates strength in overall mental toughness. Scores of 23-25 indicates average to moderate skill in mental toughness. Scores of 22 or below mean that you need to start putting more time into mental training area. Stress was assessed with the help of Everly and Girdano stress scale. This scale consists of 14 questions with four option and score ranges from 0 to 42. Anxiety was measured through Rainer Martens (1990) Sports Competition Anxiety Test. This questionnaire is consisted 15 statements with the score ranges from 11 to 33. The score less than 17 indicating low level anxiety, a score between 17 and 24 indicating the average level of anxiety and score more than 24 indicating a high level of anxiety. Self-confidence was measured by Hardy Nelson Self-confidence Questionnaire.

This question consists 56 statements with yes or no option and total score was from 0 to 56. Higher the score higher the self-confidence.

Statistical Procedure

Analysis of Covariance (ANCOVA) was used as a statistical tool

to determine the significant difference, if any existing between pre and post test data on selected variable of mental toughness, self-confidence, stress and anxiety. The level of significance was fixed at 0.05 levels.

RESULTS & DISCUSSION

Table-1 : Analysis of Covariance for Pre, Post and Adjusted mean on Mental Toughness of Experimental and Control Group

TEST	Group		S V	Sum of Squares	df	Mean Square	F ratio
	Con.	Exp.					
Pre test Mean	18.86	19.06	B	0.3	1	0.3	0.048
			W	172.666	28	6.166	
Post test Mean	19.53	22.26	B	56.033	1	56.033	8.22*
			W	190.666	28	6.809	
Adjusted Mean	19.60	22.19	B	49.997	1	49.996	14.27*
			W	94.589	27	3.503	

*Significant at 0.05 level of confidence

It was observed from the Table 1 that there was no significant difference in the pretest (0.048). The significant differences were observed in posttest (8.22) for df1 and 28 and adjusted posttest (14.27) for df1 and 27 at 0.05

level of confidence. The result of the study indicated that there was a significant improvement on mental toughness due to eight weeks relaxation practice among athletes.

Table-2 : Analysis of Covariance for Pre, Post and Adjusted mean on Self-Confidence of Experimental and Control Group

TEST	Group		S V	Sum of Squares	df	Mean Square	F ratio
	Con.	Exp.					
Pre test Mean	33.8	33.53	B	0.533	1	0.533	0.06
			W	224.133	28	8.004	
Post test Mean	34.46	40.86	B	307.2	1	307.2	27.26*
			W	315.466	28	11.266	
Adjusted Mean	34.40	40.93	B	319.304	1	319.303	33.15*
			W	260.032	27	9.630	

*Significant at 0.05 level of confidence

It was observed from the Table 2 that there was no significant

difference in the pretest (0.06). The significant differences were observed

in posttest (27.26) for df1 and 28 and adjusted posttest (33.15) for df 1 and 27 with 0.05 level of confidence. The result of the study indicated that there

was a significant improvement on self-confidence due to eight weeks relaxation practice among athletes.

Table-3: Analysis of Covariance for Pre, Post and Adjusted mean on Stress of Experimental and Control Group

TEST	Group		S V	Sum of Squares	df	Mean Square	F ratio
	Con.	Exp.					
Pre test Mean	29.6	28.46	B	9.633	1	9.633	-0.046
			W	-5760.16	28	-205.72	
Post test Mean	28.2	24.73	B	90.133	1	90.133	5.17*
			W	487.333	28	17.404	
Adjusted Mean	28.22	24.71	B	92.668	1	92.667	5.03*
			W	496.602	27	18.392	

*Significant at 0.05 level of confidence

It was observed from the Table 3 that there was no significant difference in the pretest (0.046). The significant differences were observed in posttest (5.17) for df1 and 28 and adjusted posttest (5.03) for df 1 and 27 at 0.05

level of confidence. The result of the study indicated that there was a significant reduction on stress due to eight weeks relaxation practice among athletes

Table-4: Analysis of Covariance for Pre, Post and Adjusted mean on Sports Competition Anxiety of Experimental and Control Group

TEST	Group		SV	Sum of Squares	df	Mean Square	F ratio
	Con.	Exp.					
Pre test Mean	19.66	20.06	B	1.2	1	1.2	0.19
			W	174.2667	28	6.223	
Post test Mean	19.6	16.93	B	53.33333	1	53.333	-5.71*
			W	-261.467	28	-9.338	
Adjusted Mean	19.79	16.74	B	69.298	1	69.297	-4.43*
			W	421.631	27	-15.616	

*Significant at 0.05 level of confidence

It was observed from the Table 4 that there was no significant difference in the pretest (0.19). The significant differences were observed posttest (5.71) for df1 and 28 and adjusted posttest (4.43) for df 1 and 27 at 0.05

level of confidence. The result of the study indicated that there was a significant reduction on sports competition anxiety due to eight weeks relaxation practice among athletes.

CONCLUSIONS

From the analysis of results and discussions of the present study, the following conclusions were drawn

- Due to combined relaxation training of progressive deep muscle relaxation, autogenic training and meditation overall mental toughness and self-confidence of the men athletes significantly improved.
- Due to combined relaxation training

the stress and sports competition anxiety level of the athletes significantly decreased.

- Further the result of the study indicated that relaxation training can be included as one of the training schedule in the training programme
- Further it proved that relaxation training improved the mental state of athletes, which ultimately brings positive results in the performance.

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